§2-9 The Derivative As a Function

Homework：5,9,11,23,27

Notations： $f'(x) = y' = \dfrac{dy}{dx} = \dfrac{df}{dx} = Df(x) = D_x f(x)$.

Example 1： $f(x) = x^3 - x$. Compute $f'(x)$ by definition.

Solution：

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(x+h)^3 - (x+h) - x^3 + x}{h} \\ &= 3x - 1 \end{aligned}$$

Example 2： $f(x) = \sqrt{x-1}$. Compute $f'(x)$ by definition.

Solution：

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\sqrt{x+h-1} - \sqrt{x-1}}{h} \\ &= \lim_{h\to 0} \frac{(x+h-1) - (x-1)}{h\left(\sqrt{x+h-1}\right) + \sqrt{x-1}} \\ &= \lim_{h\to 0} \frac{1}{h\left(\sqrt{x+h-1}\right) + \sqrt{x-1}} \\ &= \frac{1}{2\sqrt{x-1}}. \end{aligned}$$

Example 3： $f(x) = x|x|$. Does f has a derivative at $x = 0$?

Solution： $f'(0) = \lim_{x\to 0} \dfrac{x|x|}{x} = \lim_{x\to 0} |x| = 0$